Formas indeterminadas : $0/0$

$$\lim_{x\to 0} \frac{3x^2-13x+4}{2x^2-7x-4} = \frac{\cancel{3(0)^2}-\cancel{13(0)}+4}{\cancel{2(0)^2}-\cancel{7(0)}-4} = \frac{4}{-4} = -1$$

$$\lim_{\boxed{x\to 4}} \frac{3x^2-13x+4}{2x^2-7x-4} =$$

$\boxed{x-4}$

$$3x^2-13x+4 = (x-4)(3x-1)$$

$1x \quad -4$

$3x \quad -1$

$$2x^2-7x-4 = (x-4)(2x+1)$$

$x \quad -4$

$2x \quad +1$

$$\lim_{\boxed{x\to 4}} \frac{3x^2-13x+4}{2x^2-7x-4} = \lim_{x\to 4} \frac{\cancel{(x-4)}(3x-1)}{\cancel{(x-4)}(2x+1)} = \lim_{x\to 4} \frac{3x-1}{2x+1} = \frac{3(4)-1}{2(4)+1}$$

$$= 11/9$$

$$\lim_{x\to 0} \frac{\sqrt{4+x}-2}{x} = \lim_{x\to 0} \frac{\overset{a}{\sqrt{4+x}}-\overset{b}{2}}{x} \cdot \frac{\overset{a}{\sqrt{4+x}}+\overset{b}{2}}{\sqrt{4+x}+2}$$

$$= \lim_{x\to 0} \frac{\cancel{4}+x-\cancel{4}}{x(\sqrt{4+x}+2)} = \lim_{x\to 0} \frac{\cancel{x}}{\cancel{x}(\sqrt{4+x}+2)}$$

$$= \lim_{x\to 0} \frac{1}{\sqrt{4+x}+2} = \frac{1}{\sqrt{4+0}+2} = \frac{1}{4}$$

$$\lim_{x\to 2} \frac{x-2}{\sqrt{4x+1}-3} = \lim_{x\to 2} \frac{x-2}{\underbrace{\sqrt{4x+1}-3}_{(a-b)}} \cdot \frac{\sqrt{4x+1}+3}{\underbrace{\sqrt{4x+1}+3}_{(a+b)}}$$

$$= \lim_{x\to 2} \frac{(x-2)(\sqrt{4x+1}+3)}{4x+1-9} = \lim_{x\to 2} \frac{(x-2)(\sqrt{4x+1}+3)}{4x-8}$$

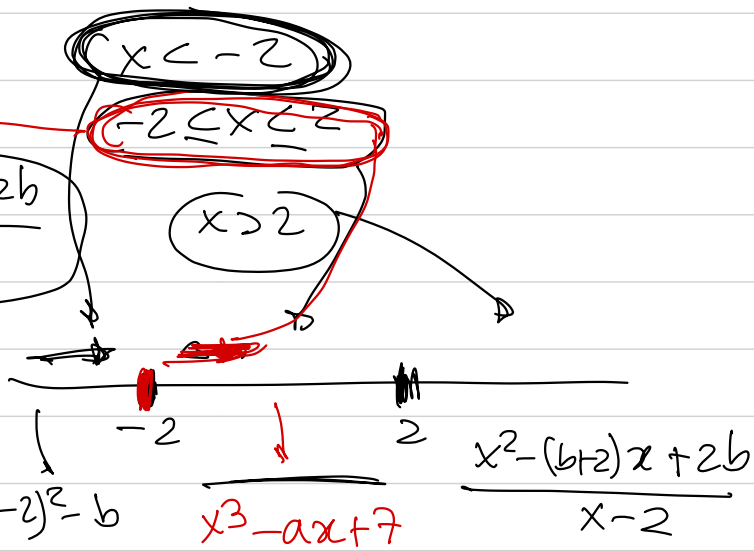
$$= \lim_{x\to 2} \frac{\cancel{(x-2)}(\sqrt{4x+1}+3)}{4\cancel{(x-2)}} = \lim_{x\to 2} \frac{\sqrt{4x+1}+3}{4}$$

$$= \frac{6}{4} = 3/2.$$

$$\boxed{\lim_{x\to 2} \frac{x-2}{\sqrt{4x+1}+3} = \frac{0}{6} = 0}$$

Limites Laterais

$$f(x) = \begin{cases} (x-2)^2 - b & x < -2 \\ x^3 - ax + 7 & -2 \le x \le 2 \\ \dfrac{x^2-(b+2)x+2b}{x-2} & x > 2 \end{cases}$$



$$\lim_{x\to -2} f(x)$$

$$\lim_{x\to -2^+} f(x) = \lim_{x\to -2^+} x^3 - ax + 7 = (-2)^3 - a(-2) + 7$$

$$= -8 + 2a + 7 = 2a - 1$$

$$\lim_{x \to -2^-} f(x) = \lim_{x \to -2^-} (x-2)^2 - b = (-2-2)^2 - b = 16 - b$$

$\lim_{x \to -2} f(x)$ ele existe se e somente se $2a - 1 = 16 - b$

$$\boxed{2a + b = 17}$$

$$\lim_{x \to 2} f(x)$$

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} x^3 - ax + 7 = 2^3 - 2a + 7 = 15 - 2a$$

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} \frac{x^2 - (b+2)x + 2b}{x - 2} = \lim_{x \to 2} \frac{\cancel{(x-2)}(x-b)}{\cancel{x-2}}$$

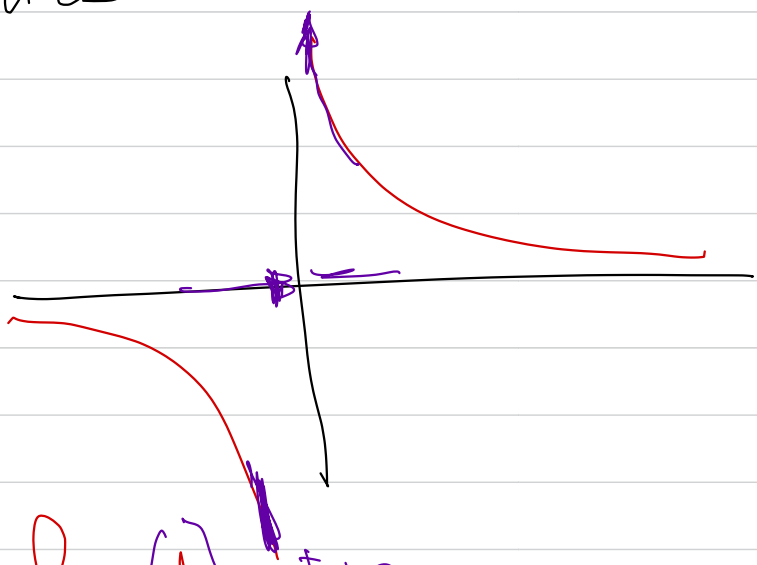
$$= \lim_{x \to 2} x - b = 2 - b$$

$\lim_{x \to 2} f(x)$ existe se e só se, $2 - b = 15 - 2a$

$$\boxed{2a - b = 13}$$

Limites infinitos

$f(x) = \frac{1}{x}$

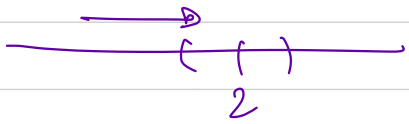


$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{1}{x} = +\infty$$

$$\lim_{x \to 0^-} f(x) = \boxed{\lim_{x \to 0^-} \frac{1}{x} = -\infty}$$

$$\lim_{x \to 2^-} \boxed{\frac{x}{x-2}}$$

$\frac{2}{0}$ indeterminação.



$x \to 2^-$

$x < 2$

$x - 2 < 0$

$0 < 1 < x < 2$

$x > 0$

$x - 2 < 0$ $\Rightarrow \boxed{\frac{x}{x-2}} < 0$

$$\lim_{x \to 2^-} \frac{x}{x-2} = -\infty$$

$$\lim_{x\to 2^-} \frac{x}{\boxed{x-2}} \quad y = x-2 \to x = y+2$$

$$x \to 2^- \Rightarrow y = x-2 \to 0^-$$

$$\lim_{y\to 0^-} \frac{y+2}{y} = \underbrace{\boxed{\lim_{y\to 0^-}(y+2)}}_{(2)} \underbrace{\boxed{\lim_{y\to 0^-}\frac{1}{y}}}_{(-\infty)} = -\infty.$$

$$\lim_{x\to 1^-} \frac{1}{x^2-1} \qquad \frac{1}{0}$$

$$y = x^2-1 = \underbrace{(x-1)}_{(-)}\underbrace{(x+1)}_{(+)}$$

Logo se $x \to 1^- \Rightarrow x < 1 \Rightarrow x-1 < 0$

Daí, $\boxed{\frac{1}{(x-1)(x+1)}} < 0$ Logo $\lim_{x\to 1^-} \frac{1}{x^2-1} = -\infty$

$$y = x^2-1 = (x-1)(x+1)$$

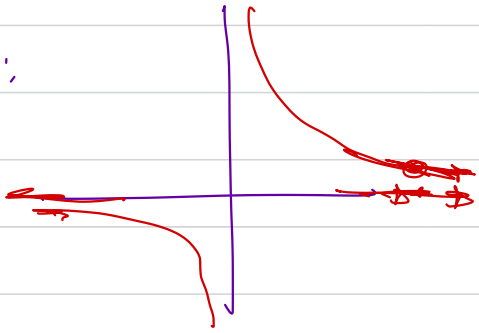
$$x \to 1^- \Rightarrow y \to 0^-$$

$$\lim_{x\to 1^-} \frac{1}{x^2-1} = \lim_{y\to 0^-} \frac{1}{y} = -\infty.$$

$\frac{0}{0}$ precisa de manipulações algébricas

$f(x) \to \frac{c}{0}$ vocês precisam estudar o sinal da f p/, $f>0 \Rightarrow L=\infty$, $f<0 \Rightarrow L=-\infty$

Limites no infinito:



$$\lim_{x\to\infty} \frac{1}{x} = 0$$

$$\lim_{x\to-\infty} \frac{1}{x} = 0$$

Ex: $\lim_{x\to\infty} \frac{x+3}{x-2}$ $\frac{\infty}{\infty}$ indeterminação.

$$= \lim_{x\to\infty} \frac{\cancel{x}(1+3/x)}{\cancel{x}(1-2/x)} = \lim_{x\to\infty} \frac{1+\underbrace{3/x}_{\to 0}}{1-\underbrace{2/x}_{\to 0}} = \frac{1}{1} = 1$$

$$\lim_{x\to\infty} \frac{3x^4+2x^2-4x}{x^5+2x^3} = \lim_{x\to\infty} \frac{\cancel{x^4}(3+2/x^2-4/x^3)}{\cancel{x^5}(1+2/x^2)}$$

x

$$= \lim_{x\to\infty} \frac{1}{x} \cdot \boxed{\lim_{x\to\infty} \frac{3+\overbrace{2/x^2}^{\to 0}-\overbrace{4/x^3}^{\to 0}}{1+\underbrace{2/x^2}_{\to}}} = 0\cdot\left(\frac{3}{1}\right) = 0$$

$$|x| = \begin{cases} x & x>0 \\ -x, & x<0 \end{cases}$$

$$\lim_{x\to-\infty} \frac{|x|}{x^2+1} = \lim_{x\to-\infty} \frac{-x}{x^2+1} = \lim_{x\to-\infty} \frac{\cancel{x}^{1}(-1)}{\cancel{x^2}_{x}(1+1/x^2)}$$

$$= \lim_{x\to-\infty} \frac{1}{x} \cdot \boxed{\lim_{x\to\infty} \frac{-1}{1+\underbrace{1/x^2}_{\to 0}}}$$

$$= 0\left(\frac{-1}{1}\right) = 0.$$

$$\lim_{x\to-\infty} \frac{4x^5+1}{\sqrt{9x^{10}-2}} = \lim_{x\to-\infty} \frac{x^5(4+1/x^5)}{\sqrt{x^{10}(9-2/x^{10})}}$$

$$= \lim_{x\to-\infty} \frac{x^5}{\sqrt{x^{10}}} \boxed{\lim_{x\to-\infty} \frac{4+\underbrace{1/x^5}_{\to 0}}{\sqrt{9-\underbrace{2/x^{10}}_{\to 0}}}}$$

$$= \lim_{x\to-\infty} \frac{x^5}{|x^5|} \frac{4}{3} = \frac{4}{3} \lim_{x\to-\infty} \frac{\cancel{x^5}}{-\cancel{x^5}}$$

$$= \frac{4}{3}(-1) = -4/3$$

$$\sqrt{x^2} = |x|$$

$x^5 = y$

$$\sqrt{x^{10}} = \sqrt{(x^5)^2} = \sqrt{y^2} = |y| = |x^5|$$

$$= |x^{10/2}| = |x^5|$$

$$\sqrt{(x^2)} = |x^{2/2}| = |x|$$

$$\sqrt{x^{10}} = |x^{10/2}| = |x^5|$$

$$(\sqrt{x})^2 \neq x$$

$$f \leq g \leq h$$

$$\lim f = \lim h = L \Rightarrow \lim g = L$$

$$\lim_{x \to \infty} \frac{\cos x}{x}$$

$$f \leq g \leq h$$

$$\frac{-1}{x} \leq \frac{\cos x}{x} \leq \frac{1}{x}$$

$$\downarrow \qquad \qquad \downarrow$$

$$0 \qquad \qquad 0$$

$$\lim_{x \to \infty} f = \lim_{x \to \infty} \frac{-1}{x} = 0$$

consequentemente $\lim_{x \to \infty} \frac{\cos x}{x} = 0$

$$\lim_{x\to\infty} \frac{\operatorname{sen} x}{x^2+4}$$

$$\underset{f}{\frac{-1}{x^2+4}} \le \underset{g}{\frac{\operatorname{sen} x}{x^2+4}} \le \underset{h}{\frac{1}{x^2+4}}$$

$$\frac{-1}{x^2+4} \to 0 \qquad \frac{1}{x^2+4} \to 0$$

$$\lim_{x\to\infty} \frac{-1}{x^2+4} = 0$$ Conseq. ...

$$\lim_{x\to\infty} \frac{\operatorname{sen} x}{x^2+4} = 0.$$

$$\boxed{\lim_{x\to\infty} \frac{x+\operatorname{sen} x}{x-\cos x}} \to \text{exercício.}$$

Indeterminações $\infty - \infty$

$$\lim_{x\to\infty} \sqrt{x+2} - \sqrt{2x}$$

$$= \lim_{x\to\infty} \sqrt{x}\left(1+2/x\right) - \sqrt{x}\cdot 2$$

$$= \lim_{x\to\infty} \sqrt{x}\left(\sqrt{1+2/x} - \sqrt{2}\right)$$

$$= \lim_{x\to\infty} \sqrt{x} \cdot \lim_{x\to\infty} \left(\overbrace{\sqrt{1+\underbrace{2/x}_{\to 0}} - \sqrt{2}}^{1-\sqrt{2}}\right)$$

$$\infty \cdot (1-\sqrt{2}) = -\infty.$$

$$\lim_{x\to\infty} \sqrt{x+3} - \sqrt{x+2}$$

$$= \lim_{x\to\infty} \sqrt{x(1+3/x)} - \sqrt{x(1+2/x)}$$

$$= \left\{\lim_{x\to\infty} \sqrt{x}\right\}\left(\sqrt{1+\underbrace{3/x}_{0}} - \sqrt{1+2/x}\right)$$

$$= \infty\,(1-1) = \infty \cdot 0.$$

Isto não serve pois aparece uma nova indeterminação

$$\lim_{x\to\infty} \overset{(a}{\sqrt{x+3}} \overset{-\,b)}{-\sqrt{x+2}} \cdot \frac{\overset{(a}{\sqrt{x+3}} \overset{+\,b)}{+\sqrt{x+2}}}{\sqrt{x+3}+\sqrt{x+2}}$$

$$= \lim_{x\to\infty} \frac{(x+3)-(x+2)}{\sqrt{x+3}+\sqrt{x+2}} = \lim_{x\to\infty} \frac{\cancel{x}+3-\cancel{x}-2}{\sqrt{x+3}+\sqrt{x+2}}$$

$$= \lim_{x\to\infty} \frac{1}{\sqrt{x+3}+\sqrt{x+2}} = 0$$

$\boxed{\infty - \infty}$ indeterminação

$$\lim_{x\to\infty} x^2 - 2x$$

$$= \lim_{x\to\infty} x\,(x-2)$$

$$= \lim_{x\to\infty} x \cdot \lim_{x\to\infty} x-2$$

$$= \infty\,(\infty) = \infty$$

1^∞ indeterminacion

$$\boxed{\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x} = e}$$

$$\lim_{h\to 0}(1+h)^{1/h} = e$$

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x} = \lim_{x\to\infty}\left[\left(1+\frac{1}{x}\right)^{x}\right]^{2}$$

$$= \left[\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}\right]^{2} = e^{2}.$$

$$\lim_{x\to\infty}\left(1+\frac{3}{x}\right)^{x}$$

$$\lim_{x\to\infty}\left(1+\frac{3/3}{x/3}\right)^{x} = \lim_{x\to\infty}\left(1+\frac{1}{x/3}\right)^{x}$$

$y = x/3 \Rightarrow x = 3y$

Se $x\to\infty \Rightarrow y\to\infty$

$$\lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{3y} = \left[\lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{y}\right]^{3} = e^{3}$$

$$\lim_{x\to\infty}\left(\frac{x+3}{x+2}\right)^x = \lim_{x\to\infty}\left(\frac{\cancel{x}(1+3/x)}{\cancel{x}(1+2/x)}\right)^x$$

$$= \lim_{x\to\infty}\frac{(1+3/x)^x}{(1+2/x)^x} = \frac{\lim_{x\to\infty}(1+3/x)^x}{\lim_{x\to\infty}\left(1+\frac{2}{x}\right)^x} = \frac{e^3}{e^2} = e$$

$$\lim_{x\to\infty}\left(1+\frac{2/2}{x/2}\right)^x = \lim_{x\to\infty}\left(1+\frac{1}{x/2}\right)^x \qquad y=x/2 \quad x=2y$$

$$= \lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{2y} = \left[\lim_{y\to\infty}\left(1+\frac{1}{y}\right)^y\right]^2 = e^2$$

$$f(x)=x^2 \qquad g(x)=\frac{x+2}{4}$$

$$(f\circ g)(x) = f(g(x)) = [g(x)]^2 = \left(\frac{x+2}{4}\right)^2 \rightarrow$$

$$(g\circ f)(x) = g(f(x)) = \frac{f(x)+2}{4} = \frac{x^2+2}{4} \rightarrow$$